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Course: 26SB MIS 581
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Date: June 01, 2026

The Research Problem

The Disease: 'Diabetes' is a serious chronic disease that greatly plagues individuals each year, where individuals lose the ability to effectively regulate levels of glucose in the blood.

The Problem: There exists a significant issue regarding under and delayed diagnoses long after irreversible changes have begun to impact the human body.

The Gap: Healthcare providers often lack high accessibility to data-driven prediction models capable of flagging demographic factors prior to blood testing for chronic conditions like Diabetes.



Dataset and Tools

Dataset: 2015 Behavioral Risk Factor Surveillance System (BRFSS) dataset, consisting of 253,680 patient records and 22 multidimensional variables.

Framework: A comparative quantitative framework was utilized through SAS OnDemand for Academics.

Models: Explored Logistic Regression baseline models and advanced Random Forest machine learning.

Data Balancing: Addressed class imbalance prior to training to improve the probability of detecting minority classes (true positive cases).



Clinical Indicators (Research Question 1)

Research Question One: To what extent can lifestyle and non-invasive clinical indicators (such as BMI, Physical Activity, and Smoking status) accurately predict the likelihood of Diabetes utilizing Machine Learning?

Findings: The evidence proved to reject the Null Hypothesis.

Conclusion: Diabetes-related indicators, such as Body Mass Index, share a statistically significant predictive relationship with a diabetes diagnosis.



Socio Economic Factors (Research Question 2)

Research Question Two: How do socioeconomic related determinants (such as income level, and education) influence the predictive probability independent of an individual's physiological health?

Findings: The evidence proved to reject the Null Hypothesis.

Conclusion: Socioeconomic determinants, such as Income, share a statistically significant predictive relationship with a diabetes diagnosis.



Machine Learning Results

Evaluation: Evaluated predictive performance of both models in reference to predictive sensitivity and specificity using Confusion Matrices and overall Area Under the Curve (AUC).

Advanced Framework: Altering from traditional statistical approaches toward more advanced statistical frameworks yields better margins of accuracy.

Integration: Healthcare Machine Learning algorithms benefit from integrating social determinants identified as multidimensional to accurately and effectively flag at-risk patients prior to diagnoses.



Ethical Considerations

Algorithmic Bias: If models fail to adequately represent minority or low-income populations, the resulting algorithms will generate skewed predictions that actively exacerbate systemic healthcare disparities.

Black Boxes: AI systems often lack transparency in their decision-making pathways, and may perpetuate biases embedded in training data.

Automation Bias: Clinicians deferring too readily to algorithmic outputs raises concerns about the erosion of clinical judgement.

Dataset Limitations: Inherent limitations include self-reporting bias (telephone surveys) and a lack of complex genetic or hereditary indicators.



Recommendations

Integration into EHRs: When integrated into electronic health records (EHRs), Machine Learning tools are capable of improving clinical workflows and reducing errors in diagnoses.

Transparency: Integrating predictive tools into clinical practice requires extreme transparency to maintain patient trust; models must be interpretable and morally balanced.

Proactive Shift: Formulated recommendations emphasize a resilient and urgent shift toward healthcare practices which promote a proactive, data-driven diagnostic approach.



Conclusion

Overall Finding: Both lifestyle/clinical indicators and socioeconomic determinants statistically influence the predictive probability of a Diabetes diagnosis independently.

Data-Driven Approach: A highly effective predictive model must take a data-driven approach that incorporates multidimensional related variables to successfully identify at-risk groups.

Conclusive Thoughts: The application of Healthcare Predictive Data Analytics allows the Healthcare Sector to formulate early prevention strategies, ultimately improving the quality and proactiveness of patient care



	A	B	C	D	E	F	G	H	I	J
	Table1									
1		Variable Name	Data Type	Description						
2	Target Variable	Diabetes_012	Categorical	0 = No Diabetes, 1 = Prediabetes, 2 = Diabetes						
3	Clinical & Biometric	HighBP	Binary	0 = No High BP, 1 = High BP						
4		HighChol	Binary	0 = No High Cholesterol, 1 = High Cholesterol						
5		BMI	Continuous	Body Mass Index (calculated numeric value)						
6		GenHlth	Ordinal	Self-reported health scale (1 = Excellent to 5 = Poor)						
7		MentHlth / PhysHlth	Continuous	Days of poor mental/physical health in past 30 days						
8		DiffWalk	Binary	Difficulty walking or climbing stairs (0 = No, 1 = Yes)						
9		Stroke / HeartDisease	Binary	History of cardiovascular events (0 = No, 1 = Yes)						
10	Lifestyle Behaviors	Smoker	Binary	Smoked 100+ cigarettes in lifetime (0 = No, 1 = Yes)						
11		PhysActivity	Binary	Physical activity in past 30 days (0 = No, 1 = Yes)						
12		Fruits / Veggies	Binary	Consumes 1+ fruit/vegetable per day (0 = No, 1 = Yes)						
13		HvyAlcoholConsump	Binary	Heavy alcohol consumption (0 = No, 1 = Yes)						
14	Socioeconomic	Age	Ordinal	13-level age category (1 = 18-24 ... 13 = 80 or older)						
15		Education	Ordinal	Education level (1 = Never attended ... 6 = College graduate)						
16		Income	Ordinal	Income bracket (1 = <\$10k ... 8 = >\$75k)						
17		AnyHealthcare	Binary	Has health care coverage (0 = No, 1 = Yes)						
18		NoDocbcCost	Binary	Could not see doctor due to cost (0 = No, 1 = Yes)						

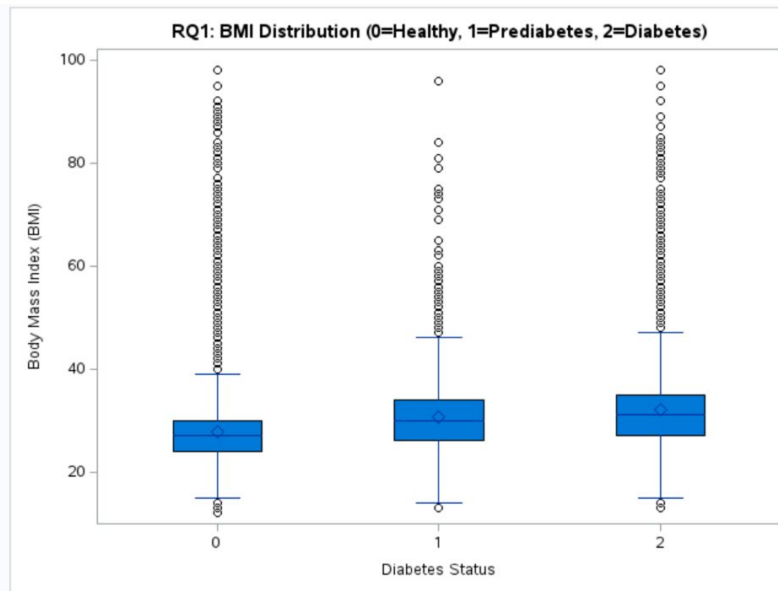
A1	fx	A	B	C	D	E	F	G
	Table1							
1		Task Description	Status	SAS Syntax / Technical Tool	Paper Section Alignment	#	Word Count Goal	Notes & Directive Alignment
2	1. Prep	Upload Dataset & Run LIBNAME	TRUE	libname mydata '...';	Methodology		200	Document the SAS environment and file
3	1. Prep	Verify Variables (PROC CONTENTS)	TRUE	proc contents data=mydata.diabetes;	Methodology		400	Detail the 22 variables; explain Binary vs.
4	1. Prep	Data Cleaning & Null Check	TRUE	proc means data=mydata.diabetes nmiss;	Methodology		400	Discuss data integrity and pre-processing
5	2. EDA	Target Variable Distribution	TRUE	proc freq; tables Diabetes_012;	Results		500	Explain the 84% class imbalance in the B
6	2. EDA	Descriptive Stats (BMI/PhysHlth)	TRUE	proc means mean std min max;	Results		400	Provide clinical baselines for the populat
7	2. EDA	Correlation Matrix	TRUE	proc corr data=mydata.diabetes;	Results		500	Identify initial relationships between risk
8	2. EDA	Socioeconomic Bar Charts	TRUE	proc sgplot; vbar Income / group=Diabetes_012;	Results		500	Visualize how Income/Education affect c
9	3. Modeling	Class Imbalance Resolution	FALSE	Weighting or selection=bal	Methodology		600	Address ethics of bias and sampling tec
10	3. Modeling	Run Multinomial Logistic Regression	FALSE	proc logistic selection=backward;	Analysis		800	Test H0 vs. H1. Detail the p-value selecti
11	3. Modeling	Generate ROC Curve & AUC	FALSE	ods graphics on;	Analysis		600	Explain model sensitivity (finding actual
12	3. Modeling	Execute Random Forest	FALSE	proc forest data=mydata.diabetes;	Analysis		600	Compare Ensemble vs. Regression perfor
13	4. Writing	Draft Formal Abstract	FALSE	APA 7 Standards	Front Matter		300	Summarize entire project scope (separat
14	4. Writing	Background & Literature Review	FALSE	6-10 Credible Sources	Section 2		1200	Synthesize 2021-2026 research on ML in
15	4. Writing	Governance & Ethics	FALSE	HIPAA / HIPAA / De-id	Section 6		800	Address PII, security, and algorithmic fai
16	5. Analysis	Interpret Odds Ratios	FALSE	SAS Output Table	Discussion		700	Quantify the real-world risk of lifestyle fa
17	5. Analysis	3 Organizational Recommendations	FALSE	Strategic Planning	Section 8		600	Actionable steps for Hospitals/Insurers.
18	Final	Draft Full Paper (5,250+ words)	FALSE	Project Write-Up Template	Full Report		5,250+	Combine all Module 5 & 6 drafts into fina

*diabetes_012_health_indicators_BRFSS2015 × Program 1 ×

CODE LOG RESULTS OUTPUT DATA



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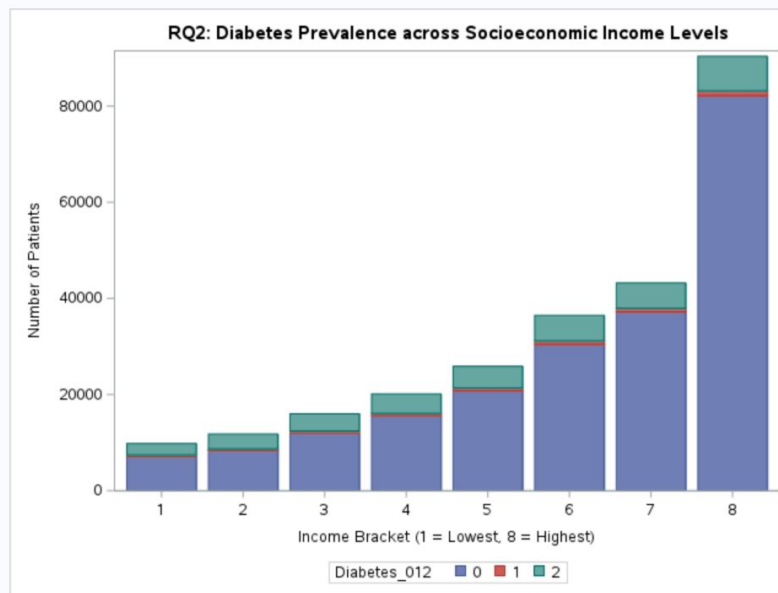
**RQ2: Diabetes Prevalence across Socioeconomic Income Levels**

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Baseline Model: Ordinal Logistic Regression

The LOGISTIC Procedure

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Probabilities modeled are cumulated over the lower Ordered Values.

Class Level Information									
Class	Value	Design Variables							
Smoker	0	0							
	1	1							
PhysActivity	0	0							
	1	1							
Education	1	0	0	0	0	0			
	2	1	0	0	0	0			
	3	0	1	0	0	0			
	4	0	0	1	0	0			
	5	0	0	0	1	0			
	6	0	0	0	0	1			
Income	1	0	0	0	0	0	0	0	
	2	1	0	0	0	0	0	0	
	3	0	1	0	0	0	0	0	
	4	0	0	1	0	0	0	0	
	5	0	0	0	1	0	0	0	
	6	0	0	0	0	1	0	0	
	7	0	0	0	0	0	1	0	
	8	0	0	0	0	0	0	1	

Model Convergence Status

Convergence criterion (GCONV=1E-8) satisfied.

Score Test for the Proportional Odds Assumption

Chi-Square	DF	Pr > ChiSq
196.5603	19	<.0001

Model Fit Statistics

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Model Fit Statistics

Criterion	Intercept Only	Intercept and Covariates
AIC	249702.94	205337.16
SC	249723.83	205556.48
-2 Log L	249698.94	205295.16

Testing Global Null Hypothesis: BETA=0

Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	44403.7824	19	<.0001
Score	41944.3067	19	<.0001
Wald	32450.9748	19	<.0001

Type 3 Analysis of Effects

Effect	DF	Wald Chi-Square	Pr > ChiSq
BMI	1	5442.0363	<.0001
HighBP	1	3040.2109	<.0001
HighChol	1	2356.0252	<.0001
Smoker	1	0.0018	0.9658
PhysActivity	1	7.2485	0.0071
Age	1	2972.4289	<.0001
Education	5	71.2682	<.0001
Income	7	238.5735	<.0001
GenHlth	1	6532.3690	<.0001

Analysis of Maximum Likelihood Estimates

Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	0	1	6.6033	0.1943	1155.2561	<.0001
Intercept	1	1	6.7791	0.1943	1217.1632	<.0001
BMI		1	-0.0628	0.000852	5442.0363	<.0001

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Baseline Fit Statistics	
Statistic	Value
Average Square Error	0.090
Misclassification Rate	0.158
Log Loss	0.492

Fit Statistics							
Number of Trees	Number of Leaves	Average Square Error (Train)	Average Square Error (OOB)	Misclassification Rate (Train)	Misclassification Rate (OOB)	Log Loss (Train)	Log Loss (OOB)
1	8909	0.0726	0.0858	0.146	0.163	0.655	1.149
2	18233	0.0677	0.0846	0.140	0.163	0.393	1.038
3	26526	0.0664	0.0832	0.138	0.162	0.353	0.920
4	33241	0.0660	0.0818	0.137	0.161	0.344	0.798
5	41604	0.0655	0.0808	0.137	0.160	0.339	0.721
6	48931	0.0654	0.0799	0.137	0.159	0.338	0.655
7	55616	0.0655	0.0792	0.137	0.158	0.338	0.605
8	64121	0.0653	0.0787	0.137	0.157	0.336	0.571
9	71553	0.0652	0.0782	0.137	0.157	0.336	0.539
10	80583	0.0650	0.0779	0.136	0.157	0.335	0.516
11	87946	0.0649	0.0776	0.136	0.156	0.335	0.494
12	95338	0.0649	0.0773	0.136	0.156	0.334	0.480
13	102965	0.0649	0.0771	0.136	0.156	0.334	0.468
14	111029	0.0648	0.0770	0.136	0.155	0.334	0.459
15	119438	0.0647	0.0768	0.136	0.155	0.333	0.452
16	127426	0.0647	0.0767	0.136	0.155	0.333	0.446
17	136309	0.0646	0.0766	0.136	0.155	0.333	0.441
18	144232	0.0646	0.0765	0.136	0.155	0.333	0.436
19	151485	0.0646	0.0763	0.136	0.154	0.333	0.432
20	159197	0.0646	0.0763	0.136	0.154	0.333	0.430

References

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